Power Grid Vulnerability Assessment against Wildfires Using Probabilistic Progression Estimation Model

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***Abstract*—Wildfires are natural or manmade disaster events that occur in forested and grassland ecosystems. Power transmission lines passing through these areas are in particular vulnerable to wildfires. The heat generated by the fire can increase the surface temperature of the overhead conductors in its vicinity. Extreme surface temperatures can lead to conductor annealing and permanent loss of tensile strength. To avoid this, the thermal rating of the at-risk lines should be dynamically adjusted, so that reduced loading of the line counteract the heat gained from the fire. However, this can affect the flow of power and dispatch of the generation units. The goal of this paper is to analyze the impact of a wildfire on the operation of a power grid. Presence of wind has a major effect on the behavior of the fire and its impact on the line. Higher winds cool down the conductor through convective heat transfer, but at the same time, move the fire front closer to the line, leading to exposure to more heat. Due to the randomness in wind speed and direction, the study presented here is based on probabilistic optimal power flow.**

***Index Terms*—Natural disaster, optimal power flow, power grid resilience, probabilistic OPF, wildfire.**

I. Nomenclature

A'	Projected area of conductor per unit length [m^2/m]
D	Conductor diameter [mm]
h_c	Convective heat transfer coefficient [W/m^2]
h_M	Vertical position of conductor [m]
$I_{C,n}$	Current flow through the conductor under normal condition [A]
$I_{C,f}$	Current flow through the conductor when wildfire is approaching the line [A]
k_a	Thermal conductivity of air [W/m°-C]
K_w	Wind direction with respect to the conductor axis [degree]
L	Flame length [m]
P_{gi}	Active power output of generating unit *i* [MW]
$P_{gi,0}$	Active power output of generating unit *i* under normal condition [MW]
$P_{gi,max}$	Rated capacity of generating unit *i* [MW]
Q_c	Convective heat loss rate per unit length of the conductor [W/m]
$Q_{c,low}$	Forced convective heat loss rate per unit length of the conductor for low wind speed [W/m]
$Q_{c,high}$	Forced convective heat loss rate per unit length of the conductor for high wind speed [W/m]
$Q_{c,n}$	Natural convective heat loss rate per unit length of the conductor [W/m]
Q_r	Radiative heat loss rate per unit length of the conductor [W/m]
$Q_{r,f}$	Radiative heat transfer rate of the fire per unit length of the conductor [W/m]
Q_s	Solar irradiance heat transfer rate per unit length of the conductor [W/m]
Q_{se}	Total solar and sky radiated heat flux rate (elevation corrected) [W/m^2]
R	Conductor resistance [Ω/m]
Res_t	Total reserve requirement at time *t* [MW]
r_{gi}	Reserve allocation amount for generator *i* [MW]
$r_{gi,max}$	Maximum reserve capacity of generator *i* [MW]
S_{max}	Conductor rating/capacity/ampacity [MVA]
t	Time [s, hr]
T_a	Ambient air temperature [°C]
T_c	Conductor's surface temperature [°C]
V_f	Rate of spread of the fire [m/s]
V_w	Wind speed [m/s]
W	Flame width [m]
α	Solar absorptivity [dimensionless]
γ	Flame tilt angle [degree]
Δ_{gi}	Physical ramp rate of generator *i* [MW/min]
ε_{FZ}	Flame zone emissivity [dimensionless]
ε_C	Conductor emissivity [dimensionless]
θ_{FZ}	Flame zone temperature [K]
θ_s	Effective angle of solar irradiance [degree]
μ_a	Dynamic viscosity of air [Pa-s]
ρ_a	Density of air [kg/m^3]
ρ_b	Fuel bulk density [kg/m^3]
σ	Stefan-Boltzmann constant [W/m^2-K^4]
τ	Atmospheric transmissivity [dimensionless]
Φ_R	Radiative heat flux of the fire front [W/m^2]

II. Introduction

Power systems are vulnerable to large-scale disturbances such as natural disasters [1]. These events may push the power grid toward its operational limits, leading to insecure

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operation, or in extreme cases, blackouts. Wildfires are one of the most common natural disasters in forested and grassland ecosystems. Since power lines in these regions may be built near fire-prone areas, wildfires can be a real threat to the network. Large fires can damage towers and poles leading to potential collapse of the line [2]. In addition, when a wildfire approaches an overhead line, the surface temperature of the conductor can increase as a result of the heat released by the fire. This can affect the conductor's rate of annealing and reduce its tensile strength [3] and/or lead to the violation of safety clearance between the conductor and the ground due to excess conductor sag. This temperature rise must be counteracted by adjusting the rating of the line, i.e. dynamic thermal rating (DTR). A change in the power transmission capacity of overhead lines can affect the dispatch of the generation units within the system [4].

In order to analyze the effect of wildfire on the network operation, an optimal power flow (OPF) model is required which considers the variations in the available capacity of the overhead lines. By ensuring the line flows are below their respective DTR values, the operator can minimize the long-term damages to the lines and, in the short term, potential instances of flashover between lines and the ground. Moreover, as the generation levels of all units may in general be affected by this modification, the reserve allocations of the units also need to be updated. Failure to do this, could lead to possible load shedding under emergency conditions which is highly undesirable for electric utilities.

To model network operation under the threat of a progressing wildfire, all factors affecting the fire including but not limited to weather conditions and type of fuel must be taken into account. Among these factors, wind can be considered as one with major impact on the wildfire behavior. Wind is probabilistic in both its speed and direction. Both these parameters can significantly affect the progress behavior of the fire. Therefore, it is necessary to incorporate the probability density functions (PDF) of wind speed and wind direction into the OPF model to track all possible scenarios and derive the expected value of dispatch decisions based on all possible samples. Hence, a probabilistic OPF is needed to determine the generation output and reserve allocations of each unit considering the probabilistic behavior of the fire. This is the focal point of this paper.

III. Current-Temprature Model of a Conductor

Conductor surface temperature depends on electrical and thermal properties of the conductor material, atmospheric factors, and load current. Thermal rating of overhead conductors can be determined based on all these factors [6]-[9]. In this paper, we use the thermal balance model proposed in IEEE Standard 738 [5] to determine the DTR of a conductor. Here, two thermal components are assumed to lead to conductor heating: Joule's effect due to the conductor's resistance and solar irradiance. On the other hand, convection and radiation heat loss mechanisms assist in cooling the conductor. At steady state, this model can be simplified in the form of an algebraic equation:

$$R \cdot I_{\mathrm{C,n}}^2 + Q_s = Q_r + Q_c \tag{1}$$

where solar, radiative and convective heat transfer rates, can be calculated as follows [5]:

$$Q_s = \alpha \cdot Q_{Se} \sin(\theta_s) \cdot A' \tag{2}$$

$$Q_r = 0.0178 \cdot D\varepsilon_C \left[\left(\frac{T_C + 273}{100} \right)^4 - \left(\frac{T_a + 273}{100} \right)^4 \right] \tag{3}$$

$$Q_{c,\mathrm{low}} = \left[1.01 + 0.0372 \left(\frac{D\rho_a V_w}{\mu_a} \right)^{0.52} \right] k_a K_w (T_C - T_a) \tag{4}$$

$$Q_{c,\mathrm{high}} = \left[0.0119 \left(\frac{D\rho_a V_w}{\mu_a} \right)^{0.6} \right] k_a K_w (T_C - T_a) \tag{5}$$

$$Q_{\mathrm{c,n}} = 0.0205 \cdot \rho_a^{0.5} D^{0.75} (T_C - T_a)^{1.25} \tag{6}$$

$$Q_c = \max(Q_{c,\mathrm{low}}, Q_{c,\mathrm{high}}, Q_{c,\mathrm{n}}) \tag{7}$$

Equations (4) and (5) represent forced convection heat loss rate for low wind speeds and high wind speeds, respectively. Equation (6) is for natural convection which occurs when the wind speed is zero. It is also assumed that the altitude and azimuth of sun will remain unchanged during the time frame of analysis. In this study, wind parameters V_w and K_w follow probabilistic behavior as discussed below.

IV. Wildfire Progression Model

Predicting the progress of a wildfire is not an easy task. Numerous studies have addressed the issue of wildfire behavior modeling (see [10] for a survey). The majority of these models are developed for surface fires, where the link between fire rate of spread, flame length, fuel bed and related heat loss and transfer rates are taken into account. In this paper, we have adopted the model developed in [11], where wildfire is modeled as a radiant surface and it is assumed that the flame body has a geometrical shape that emits the radiative heat uniformly (Fig. 1). No convective heat transfer rate is considered in this paper for the wildfire.

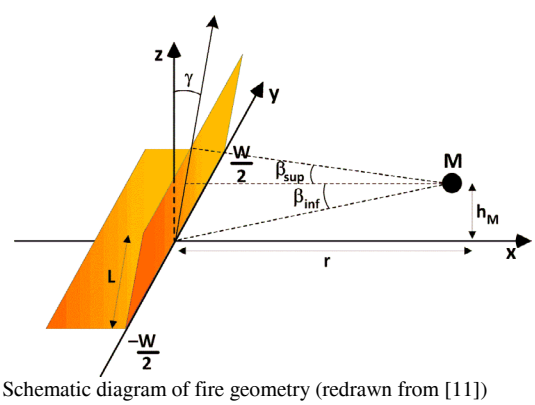

Fig. 1. Schematic diagram of fire geometry (redrawn from [11])

The radiative heat flux emitted from a flame body with length L and width W which is received by a target M located at height h_M above the ground and at distance r from the fire can be calculated as:

$$\Phi_R = \tau \cdot \varepsilon_{FZ} \sigma \cdot (\theta_{FZ})^4 \cdot f(L, h_M, r, \gamma), \tag{8}$$

where function f is provided in [11] and will not be repeated here. It is also assumed that the wildfire is moving towards the target with a specific rate of spread which is determined by the

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wind speed, wind direction and environment vegetation properties. Equation (9) is used here for wildfire rate of spread which is based on [12], where k is equal to 0.07 kg/m^2.

$$V_f = \frac{k(1+V_w)}{\rho_b} \tag{9}$$

V. Wildfire-Conductor Interaction Model

In order to evaluate the effect of a wildfire which is moving towards an overhead line on the conductor ampacity, a modified version of equation (1) has been used here based on our earlier work [4]. Since the fire causes overheating of the conductor, the wildfire related heat transfer rate should be considered in the left side of the equation:

$$R \cdot I_{\mathrm{C,f}}^2 + Q_s + Q_{r,f} = Q_r + Q_c \tag{10}$$

where:

$$Q_{r,f} = D \cdot \Phi_R \tag{11}$$

It is assumed here that the wildfire is located at an initial distance r_o from the transmission line and is moving towards the line with rate of spread in accordance with (9). Wind direction can be decomposed into two vectors, one parallel with the conductor and another perpendicular to it. Therefore, the distance between wildfire and the overhead line can be expressed as:

$$r = r_0 - V_f \cdot t \cdot \cos(K_w) \tag{12}$$

To calculate the reduction in the conductor ampacity (rating) due to wildfire heat, we find the ratio between the maximum allowable conductor current under normal operation and with wildfire (calculated from (1) and (10), respectively), and apply it to the nameplate capacity.

$$S_{\max}^{\text{new}} = \frac{I_{C,f}}{I_{C,n}} S_{\max}^{\text{old}} \tag{13}$$

VI. Prpoposed Probabilistic Wildfire Progression Estimation Model

Due to the uncertain nature of wildfire spread, it is essential to model wind speed and wind direction probabilistically. Ignoring the interdependency of these two random variables, wind can be modeled as a mixture of two PDFs. In this paper, we have used the Rayleigh distribution function [13] for estimating wind speed. For wind direction, since data are recorded in terms of degrees from 0 to 360, direction can be considered as a circular variable [14]. Hence, von Mises distribution is an appropriate model for representing randomness in wind direction [14], [15].

$$f(V_w) = \frac{V_w}{\delta^2} e^{-V_w^2/2\delta^2}, \qquad V_W \geq 0 \tag{14}$$

$$f(K_w | \mu, k) = \frac{e^{k\cos(K_w - \mu)}}{2\pi \cdot I_0 k} \tag{15}$$

where μ and k are the mean value and concentration parameters, respectively, and I_0 is the modified Bessel function of order 0. Figure 2 illustrates the histogram of wind speed variations using Rayleigh distribution by applying Monte Carlo simulation with N = 900 samples. Also, the histogram of wind direction has been shown in Fig. 3 based on Monte Carlo sampling with N = 900.

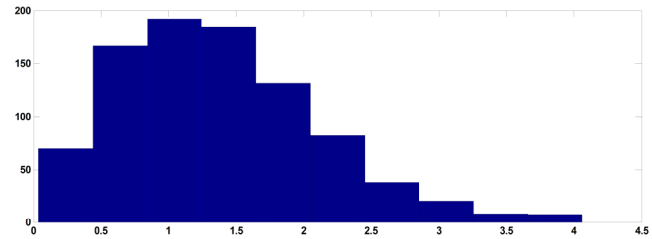

Fig. 2. Wind speed histogram

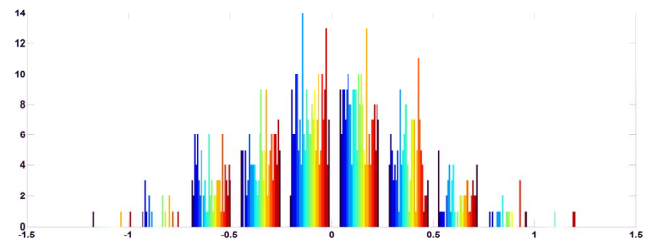

Fig. 3. Wind direction histogram

VII. Probabilistic OPF Model

A. General Approach

Monte Carlo sampling method has been used to generate random wind scenarios (based on random samples from the wind speed and wind direction PDFs). For each sample, the fire distance to the overhead line is calculated as in (12) over several time periods. Each distance value is then used to calculate the radiative heat flux generated by the fire using (8), which then provides the radiative heat transfer rate of the wildfire based on (11). Equations (1) and (10) are then solved to find the maximum allowable current flow for normal operation and for operation under wildfire. Finally, these values provide the updated DTR (from (13)). The OPF problem is then solved using the updated DTR. Each random scenario results in a single solution for the OPF problem. After performing the Monte Carlo simulation, the OPF solution (i.e. generation dispatch and reserve allocation for different units) is also determined probabilistically.

B. Reserve Allocation

The amount of reserve provided by each generator will be limited by a pre-defined upper bound $r_{gi,\max}$ as well as the physical ramp rate Δ_{gi}:

$$\forall i, \forall t : 0 \leq r_{gi,t} \leq \min(r_{gi,\max}, \Delta_{gi} \cdot d) \tag{16}$$

where d is the time window over which the entire reserve should be available (usually in the range of a few minutes). Naturally, the amount of reserve allocated to each unit plus the actual generation level must be less than the total capacity of the generator.

$$\forall i, \forall t : P_{gi,t} + r_{gi,t} \leq P_{gi,\max} \tag{17}$$

Wildfire causes certain generators to reduce the power generation levels, despite the fact that generation capacity is available and the unit may be economical. This is due to the fact that DTR of the lines leaving that generator have reduced

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as a result of the heat released by the fire. Therefore, although the generator has surplus capacity, it may not have means to provide this capacity in the form of generation or reserves to other parts of the system. To model this phenomenon, we have assumed that whenever the generation output of a unit decreases more than a specific threshold due to wildfire progression, the related reserve will be capped at 10 percent of the current generation of that unit. This is to ensure that the OPF solution does not allocate extra reserves to generation units that are unable to inject the power when necessary. This constraint has been modeled as in (18).

$$r_{gi,t} \le 0.1 P_{gi,t} \quad \text{if} \quad P_{gi,t} \le 0.95 P_{gi,0} \tag{18}$$

At each point in time, the sum of spinning reserves allocated to various generation units must meet the required total reserves in the system:

$$\forall t : \sum r_{gi,t} \ge \text{Res}_t \tag{19}$$

Finally, the cost of spinning reserve is considered to be a linear function which is determined based on the marginal cost of reserve for each generator. The operational costs of the generators are assumed to follow a second order polynomial based on the amount of generation. The problem is therefore formulated as:

$$\min \sum_{i=1}^{n} (a_i + b_i P_{gi,t} + c_i P_{gi,t}^2) + \sum_{i=1}^{n} c_i^{res} \cdot r_{gi,t} \tag{20}$$

Subject to:

- AC load flow equations,
- Node voltage constraints,
- Modified capacity constraints for overhead lines (according to (13)),
- Generator capacity constraints,
- Reserve constraints (16)-(19).

VIII. Simulation Results

In this section, the proposed probabilistic wildfire progression estimation model has been studied on the IEEE 30-bus test power system in order to evaluate the effect of a wildfire on generation outputs and reserve amounts of each generating unit. The schematic diagram of this network has been shown in Fig. 4. The ambient air temperature is assumed to be fixed at 40°C for all time steps and the density of air has been considered 1.225 kg/m^3. All other simulation parameters have been chosen based on the authors' earlier work in [4]. Reserve requirement has been assumed to be 15% of the total demand. It is assumed here that a wildfire is approaching line 23-15 and the initial distance r_o between the fire and the line is 30m. Figure 5 depicts the histogram of conductor ampacity while the wildfire is progressing towards the line in three 15min intervals. This reduction in conductor ampacity can alter the power flow of all other lines in the network and hence it can change all the generation and reserve dispatch decisions. Software package MATPOWER 5.0 has been used to perform the OPF [16].

Figs. 6 and 7 show the histograms of generation output of units 5 and 6, respectively. As the wildfire approaches line 23-15, the capacity of this line starts to decrease; therefore, it causes a reduction in the output of unit 5. Unit 6 tries to compensate for the shortage of power, as it is the only other unit in area 2 and farther away from the affected line compared to unit 5. Table 1 indicates the expected values of the generated power of other units over the three 15min time intervals.

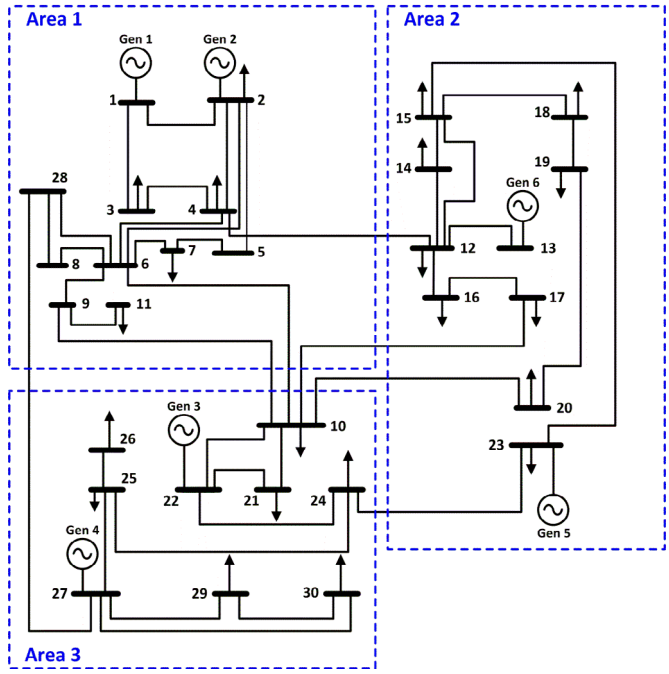

Fig. 4. Schematic diagram of the test distribution system

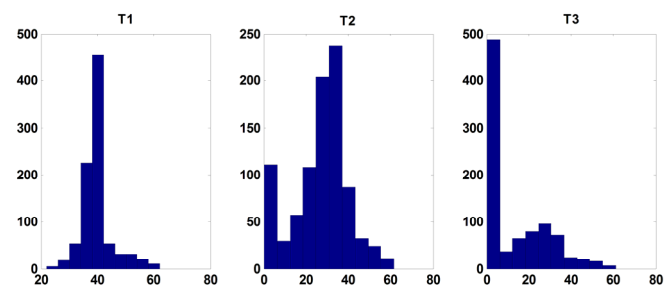

Fig. 5. Conductor ampacity variation for three 15min time intervals

As the output of unit 5 decreases, according to constraint (18), it is expected that the unit will have less participation in meeting the demand and the reserves due to the transmission line being affected. Figure 8 illustrates the kernel density estimates of the reserve allocations for unit 5, and shows how these values decrease over time. At the same time, since unit 6 has to increase its generation level in order to compensate for the shortage of power by unit 5, it gets closer to its maximum power capacity; hence, its reserve allocation also reduces. Simulations indicate that in this case unit 3 which is the next closest unit to unit 5 tries to compensate for the major part of reserve shortage in the system (see Fig. 9). Also, Table 1 indicates that the units in area 3 act as a valuable resource during the course of the wildfire. This is due to the stronger connection between areas 2 and 3 (through 3 transmission lines) compared to that of areas of 1 and 2 (only one connected line).

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TABLE I. EXPECTED VALUE OF GENERATION OUTPUTS AND RESERVE ALLOCATIONS OVER THE THREE TIME INTERVALS

Unit No.	Expected Value of Generation			Expected Value of Reserves		
	T1	T2	T3	T1	T2	T3
1	41.89	39.92	36.93	6.03	5.35	3.45
2	55.78	53.22	49.25	5.06	5.05	3.77
3	22.83	23.85	25.85	5.04	8.18	11.99
4	40.28	41.17	42.16	3.84	5.44	7.02
5	13.49	9.25	4.69	4.11	0.65	0.33
6	17.83	24.58	32.99	4.29	3.70	1.82

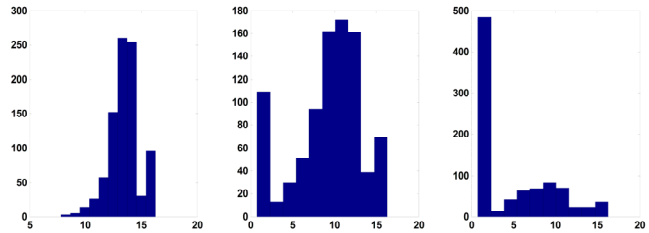

Fig. 6. Histogram of Gen 5 generation for three 15min time intervals

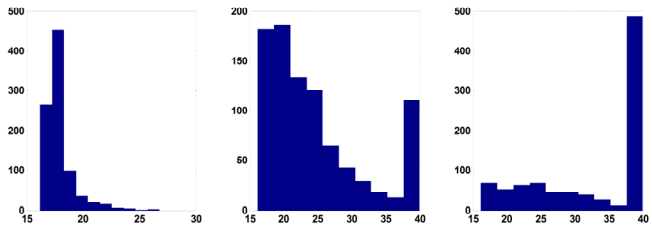

Fig. 7. Histogram of Gen 6 generation for three 15min time intervals

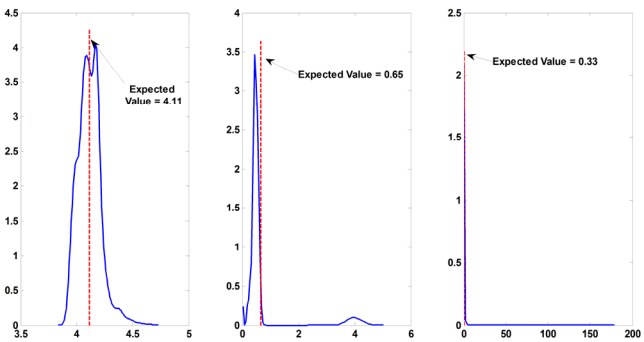

Fig. 8. Kernel estimation of reserve allocation for Gen. 5 for 3 time intervals.

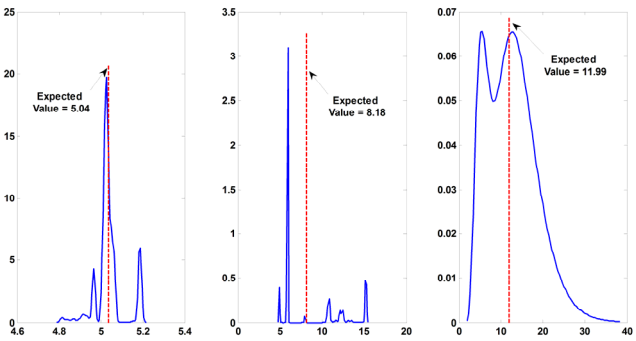

Fig. 9. Kernel estimation of reserve allocation for Gen. 3 for 3 time intervals.

IX. CONCLUSIONS

The heat flux radiated by a wildfire can affect the current carrying capacity of overhead lines in the vicinity of the fire. This can change the dispatch of the system and the amount of reserves allocated to each unit. Presence of wind can significantly affect the progression of the fire and its impact on the lines. What makes this a stochastic study, is the inherent uncertainty in wind speed and direction. A model was developed in this paper to probabilistically show the impact of wildfire on the dispatch of a power system.

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